

Estimating the Size of a DAG by Following Random Paths—Extensive Tables of Experimental Results

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Permutations										
m	$n = 7$			$n = 8$			$n = 9$			$n = 10$
	mean $\pm$ std. dev.	max.	$\hat{\chi}$	mean $\pm$ std. dev.	max.	$\hat{\chi}$	mean $\pm$ std. dev.	max.	$\hat{\chi}$	mean $\pm$ std. dev.
10^1	4,122.9 $\pm$ 9,304.0	3.03×10^4	0.73462	39,331.8 $\pm$ 59,061.9	1.42×10^5	0.36044	48,168.7 $\pm$ 56,348.3	1.68×10^5	0.34803	41,437.8 $\pm$ 37,261.1
10^2	3,444.5 $\pm$ 6,008.7	3.35×10^4	0.09723	28,772.5 $\pm$ 64,855.0	3.42×10^5	0.11875	134,918.1 $\pm$ 4.87 $\times 10^5$	4.21×10^6	0.31195	404,610.8 $\pm$ 1.29 $\times 10^6$
10^3	4,558.2 $\pm$ 14,350.9	1.95×10^5	0.04285	77,136.0 $\pm$ 1.55 $\times 10^6$	4.83×10^7	0.62640	188,542.8 $\pm$ 1.33 $\times 10^6$	2.93×10^7	0.15514	7,137,705.1 $\pm$ 1.99 $\times 10^8$
10^4	5,352.3 $\pm$ 27,948.9	1.29×10^6	0.02416	44,629.3 $\pm$ 6.57 $\times 10^5$	4.83×10^7	0.10826	221,679.5 $\pm$ 2.23 $\times 10^6$	1.49×10^8	0.06720	2,335,438.1 $\pm$ 6.86 $\times 10^7$
10^5	5,131.4 $\pm$ 29,457.4	3.51×10^6	0.00685	42,382.8 $\pm$ 1.13 $\times 10^6$	2.89×10^8	0.06821	372,754.8 $\pm$ 1.75 $\times 10^7$	4.94×10^9	0.13250	2,920,867.5 $\pm$ 1.47 $\times 10^8$
10^6	5,016.9 $\pm$ 40,230.5	2.23×10^7	0.00445	39,837.5 $\pm$ 8.32 $\times 10^5$	3.63×10^8	0.00912	347,645.5 $\pm$ 1.23 $\times 10^7$	4.98×10^9	0.01433	3,016,115.1 $\pm$ 2.69 $\times 10^8$
10^7	5,046.5 $\pm$ 38,894.2	3.39×10^7	0.00067	40,820.6 $\pm$ 1.92 $\times 10^6$	5.16×10^9	0.01265	351,350.7 $\pm$ 2.59 $\times 10^7$	4.64×10^{10}	0.01320	3,281,363.3 $\pm$ 6.06 $\times 10^8$
	5,040 $\pm$ 3.730 $\times 10^4$	#visited = 22		40,320 $\pm$ 1.642 $\times 10^6$	#visited = 29		362,880 $\pm$ 1.614 $\times 10^8$	#visited = 37		3,628,800 $\pm$ 3.682 $\times 10^{10}$

Permutations										
x.	$\hat{\chi}$	$n = 10$			$n = 11$			$n = 12$		
		mean $\pm$ std. dev.	max.	$\hat{\chi}$	mean $\pm$ std. dev.	max.	$\hat{\chi}$	mean $\pm$ std. dev.	max.	$\hat{\chi}$
10^5	0.34803	41,437.8 $\pm$ 37,261.1	1.18×10^5	0.28431	125,617.4 $\pm$ 1.37 $\times 10^5$	3.78×10^5	0.30120	3,371,039.7 $\pm$ 8.99 $\times 10^6$	2.89×10^7	0.85746
10^6	0.31195	404,610.8 $\pm$ 1.29 $\times 10^6$	9.64×10^6	0.23824	11,415,775.6 $\pm$ 5.68 $\times 10^7$	5.10×10^8	0.44649	8,845,375.4 $\pm$ 5.62 $\times 10^7$	5.59×10^8	0.63242
10^7	0.15514	7,137,705.1 $\pm$ 1.99 $\times 10^8$	6.29×10^9	0.88158	4,892,118.2 $\pm$ 2.66 $\times 10^7$	5.10×10^8	0.10419	46,067,851.4 $\pm$ 7.90 $\times 10^8$	2.39×10^{10}	0.51909
10^8	0.06720	2,335,438.1 $\pm$ 6.86 $\times 10^7$	6.29×10^9	0.26943	19,442,895.5 $\pm$ 7.83 $\times 10^8$	6.14×10^{10}	0.31570	78,356,893.3 $\pm$ 1.84 $\times 10^9$	1.33×10^{11}	0.16929
10^9	0.13250	2,920,867.5 $\pm$ 1.47 $\times 10^8$	2.87×10^{10}	0.09840	24,935,704.5 $\pm$ 2.29 $\times 10^9$	6.80×10^{11}	0.27286	112,220,287.8 $\pm$ 7.76 $\times 10^9$	2.15×10^{12}	0.19147
10^9	0.01433	3,016,115.1 $\pm$ 2.69 $\times 10^8$	1.73×10^{11}	0.05721	22,721,429.1 $\pm$ 2.33 $\times 10^9$	1.86×10^{12}	0.08186	302,925,208.2 $\pm$ 1.45 $\times 10^{11}$	1.43×10^{14}	0.47357
10^{10}	0.01320	3,281,363.3 $\pm$ 6.06 $\times 10^8$	1.34×10^{12}	0.04080	50,007,894.4 $\pm$ 6.80 $\times 10^{10}$	2.14×10^{14}	0.42825	304,480,693.4 $\pm$ 1.55 $\times 10^{11}$	3.25×10^{14}	0.10670
ted = 37		3,628,800 $\pm$ 3.682 $\times 10^{10}$	#visited = 46		39,916,800 $\pm$ 2.010 $\times 10^{13}$	#visited = 56		479,001,600 $\pm$ 2.720 $\times 10^{16}$	#visited = 67	

Table 1: Algorithm P, Path Sampling, permutations

Permutations										
m	$n = 7$			$n = 8$			$n = 9$			$n = 10$
	mean $\pm$ std. dev.	max.	$\hat{\chi}$	mean $\pm$ std. dev.	max.	$\hat{\chi}$	mean $\pm$ std. dev.	max.	$\hat{\chi}$	mean $\pm$ std. dev.
10^1	$2,134.2 \pm 2,895.3$	1.00×10^4	0.46900	$10,564.8 \pm 11,944.9$	3.32×10^4	0.31394	$234,362.6 \pm 4.41 \times 10^5$	1.28×10^6	0.54434	$485,847.2 \pm 1.01 \times 10^6$
10^2	$5,735.4 \pm 14,589.7$	1.30×10^5	0.22703	$15,677.9 \pm 28,749.5$	1.85×10^5	0.11808	$315,348.3 \pm 1.03 \times 10^6$	6.34×10^6	0.20111	$1,513,525.6 \pm 6.94 \times 10^6$
10^3	$5,322.5 \pm 17,623.9$	3.56×10^5	0.06698	$47,912.4 \pm 4.33 \times 10^5$	1.23×10^7	0.25653	$281,586.0 \pm 1.10 \times 10^6$	1.40×10^7	0.04980	$2,192,325.9 \pm 2.70 \times 10^7$
10^4	$5,198.4 \pm 21,658.4$	1.21×10^6	0.02331	$39,265.0 \pm 2.24 \times 10^5$	1.23×10^7	0.03130	$312,165.1 \pm 3.48 \times 10^6$	2.36×10^8	0.07569	$5,269,581.7 \pm 2.24 \times 10^8$
10^5	$4,949.1 \pm 17,630.6$	2.18×10^6	0.00440	$40,970.8 \pm 3.82 \times 10^5$	6.63×10^7	0.01619	$411,450.9 \pm 2.05 \times 10^7$	6.05×10^9	0.14698	$3,143,399.0 \pm 1.01 \times 10^8$
10^6	$5,010.6 \pm 17,626.0$	3.99×10^6	0.00080	$39,777.6 \pm 4.31 \times 10^5$	2.35×10^8	0.00592	$360,495.0 \pm 1.13 \times 10^7$	6.05×10^9	0.01678	$3,225,251.6 \pm 1.37 \times 10^8$
10^7	$5,033.0 \pm 18,375.8$	5.41×10^6	0.00011	$40,289.8 \pm 5.41 \times 10^5$	7.25×10^8	0.00180	$372,028.6 \pm 4.02 \times 10^7$	1.01×10^{11}	0.02708	$3,335,166.5 \pm 2.18 \times 10^8$
	5,040 $\pm 1.883 \times 10^4$	#visited = 22		40,320 $\pm 6.316 \times 10^5$	#visited = 29		362,880 $\pm 4.665 \times 10^7$	#visited = 37		3,628,800 $\pm 8.110 \times 10^9$

Permutations										
x.	$\hat{\chi}$	$n = 10$			$n = 11$			$n = 12$		
		mean $\pm$ std. dev.	max.	$\hat{\chi}$	mean $\pm$ std. dev.	max.	$\hat{\chi}$	mean $\pm$ std. dev.	max.	$\hat{\chi}$
10^6	0.54434	$485,847.2 \pm 1.01 \times 10^6$	3.33×10^6	0.68531	$2,753,283.9 \pm 6.23 \times 10^6$	2.02×10^7	0.73546	$5,659,381.0 \pm 1.70 \times 10^7$	5.41×10^7	0.95583
10^6	0.20111	$1,513,525.6 \pm 6.94 \times 10^6$	5.11×10^7	0.33763	$6,623,288.3 \pm 2.19 \times 10^7$	1.57×10^8	0.23716	$23,325,678.9 \pm 1.10 \times 10^8$	1.04×10^9	0.44646
10^7	0.04980	$2,192,325.9 \pm 2.70 \times 10^7$	8.20×10^8	0.37425	$19,736,008.9 \pm 2.40 \times 10^8$	5.05×10^9	0.25594	$84,668,334.9 \pm 1.34 \times 10^9$	4.16×10^{10}	0.49097
10^8	0.07569	$5,269,581.7 \pm 2.24 \times 10^8$	1.82×10^{10}	0.34561	$17,512,499.7 \pm 4.52 \times 10^8$	4.08×10^{10}	0.23326	$120,895,671.8 \pm 2.69 \times 10^9$	2.01×10^{11}	0.16634
10^9	0.14698	$3,143,399.0 \pm 1.01 \times 10^8$	1.82×10^{10}	0.05794	$22,871,948.2 \pm 1.29 \times 10^9$	3.44×10^{11}	0.15040	$219,123,728.8 \pm 2.14 \times 10^{10}$	6.46×10^{12}	0.29491
10^9	0.01678	$3,225,251.6 \pm 1.37 \times 10^8$	5.67×10^{10}	0.01757	$40,238,558.2 \pm 6.98 \times 10^9$	5.28×10^{12}	0.13113	$236,948,388.8 \pm 2.77 \times 10^{10}$	2.12×10^{13}	0.08961
10^{11}	0.02708	$3,335,166.5 \pm 2.18 \times 10^8$	2.83×10^{11}	0.00848	$30,446,498.9 \pm 3.63 \times 10^9$	5.28×10^{12}	0.01733	$326,092,714.7 \pm 1.32 \times 10^{11}$	3.19×10^{14}	0.09780
visited = 37		3,628,800 $\pm 8.110 \times 10^9$	#visited = 46		39,916,800 $\pm 3.400 \times 10^{12}$	#visited = 56		479,001,600 $\pm 3.540 \times 10^{15}$	#visited = 67	

Table 2: IS by outdegree, permutations

Permutations											
m	$n = 7$			$n = 8$			$n = 9$			$n = 10$	
	mean $\pm$ std. dev.	max.	$\hat{\chi}$	mean $\pm$ std. dev.	max.	$\hat{\chi}$	mean $\pm$ std. dev.	max.	$\hat{\chi}$	mean $\pm$ std. dev.	max.
10^1	$6,091.2 \pm 6,204.7$	2.09×10^4	0.34320	$34,023.3 \pm 41,991.4$	1.48×10^5	0.43367	$186,271.5 \pm 1.09 \times 10^5$	3.38×10^5	0.18146	$1,971,685.5 \pm 1.88 \times 10^6$	5.80×10^6
10^2	$4,910.5 \pm 3,884.3$	2.13×10^4	0.04343	$47,540.9 \pm 69,429.7$	4.77×10^5	0.10031	$277,473.9 \pm 4.23 \times 10^5$	3.41×10^6	0.12283	$3,714,925.1 \pm 7.23 \times 10^6$	4.58×10^7
10^3	$5,082.0 \pm 4,648.6$	4.29×10^4	0.00844	$39,992.2 \pm 47,453.0$	4.77×10^5	0.01192	$333,151.4 \pm 5.89 \times 10^5$	7.63×10^6	0.02289	$3,985,556.7 \pm 1.41 \times 10^7$	3.50×10^8
10^4	$5,081.4 \pm 4,883.8$	9.65×10^4	0.00190	$39,758.2 \pm 54,078.2$	1.14×10^6	0.00286	$362,520.5 \pm 1.09 \times 10^6$	7.77×10^7	0.02143	$3,672,126.7 \pm 1.50 \times 10^7$	9.17×10^8
10^5	$5,029.2 \pm 4,797.2$	1.46×10^5	0.00029	$40,183.6 \pm 59,067.6$	2.87×10^6	0.00071	$361,473.8 \pm 1.01 \times 10^6$	1.17×10^8	0.00324	$3,636,904.9 \pm 1.38 \times 10^7$	1.50×10^9
10^6	$5,039.0 \pm 4,844.3$	1.46×10^5	0.00003	$40,336.0 \pm 61,807.6$	5.89×10^6	0.00015	$363,775.1 \pm 9.37 \times 10^5$	1.71×10^8	0.00047	$3,622,834.1 \pm 1.65 \times 10^7$	7.46×10^9
	5,040 #visited: avg = 47.9, max = 61 #generated: avg = 137.5			40,320 #visited: avg = 74.3, max = 95 #generated: avg = 247.1			362,880 #visited: avg = 109.6, max = 144 #generated: avg = 414.9			3,628,800 #visited: avg = 155.6, max = 197 #generated: avg = 659.8	

Permutations											
max.	$\hat{\chi}$	$n = 10$			$n = 11$			$n = 12$			
		mean $\pm$ std. dev.	max.	$\hat{\chi}$	mean $\pm$ std. dev.	max.	$\hat{\chi}$	mean $\pm$ std. dev.	max.	$\hat{\chi}$	
8×10^5	0.18146	$1,971,685.5 \pm 1.88 \times 10^6$	5.80×10^6	0.29422	$28,221,404.6 \pm 3.44 \times 10^7$	8.91×10^7	0.31565	$96,426,424.5 \pm 1.31 \times 10^8$	3.73×10^8	0.38663	
1×10^6	0.12283	$3,714,925.1 \pm 7.23 \times 10^6$	4.58×10^7	0.12322	$51,184,212.1 \pm 1.42 \times 10^8$	1.22×10^9	0.23744	$387,283,255.9 \pm 1.07 \times 10^9$	8.95×10^9	0.23113	
3×10^6	0.02289	$3,985,556.7 \pm 1.41 \times 10^7$	3.50×10^8	0.08772	$39,109,123.2 \pm 1.19 \times 10^8$	1.86×10^9	0.04745	$387,170,456.1 \pm 1.81 \times 10^9$	4.40×10^{10}	0.11376	
7×10^7	0.02143	$3,672,126.7 \pm 1.50 \times 10^7$	9.17×10^8	0.02496	$39,652,300.5 \pm 2.09 \times 10^8$	1.34×10^{10}	0.03381	$473,614,045.3 \pm 4.62 \times 10^9$	3.17×10^{11}	0.06685	
7×10^8	0.00324	$3,636,904.9 \pm 1.38 \times 10^7$	1.50×10^9	0.00413	$39,691,293.3 \pm 3.08 \times 10^8$	5.14×10^{10}	0.01295	$471,672,107.2 \pm 5.86 \times 10^9$	1.02×10^{12}	0.02169	
1×10^8	0.00047	$3,622,834.1 \pm 1.65 \times 10^7$	7.46×10^9	0.00206	$40,605,515.0 \pm 3.51 \times 10^8$	1.39×10^{11}	0.00342	$471,736,695.3 \pm 5.84 \times 10^9$	3.05×10^{12}	0.00647	
max = 144		3,628,800 #visited: avg = 155.6, max = 197 #generated: avg = 659.8			39,916,800 #visited: avg = 213.7, max = 273 #generated: avg = 1003.2			479,001,600 #visited: avg = 285.6, max = 366 #generated: avg = 1469.2			

Table 3: Bucket Sampling, permutations

Permutations											
m	$n = 7$			$n = 8$			$n = 9$			$n = 10$	
	mean $\pm$ std. dev.	max.	$\hat{\chi}$	mean $\pm$ std. dev.	max.	$\hat{\chi}$	mean $\pm$ std. dev.	max.	$\hat{\chi}$	mean $\pm$ std. dev.	$\hat{\chi}$
10^1	5,621.8 $\pm$ 2,914.8	1.09×10^4	0.19423	37,629.0 $\pm$ 21,347.7	7.31×10^4	0.19427	335,864.8 $\pm$ 5.66×10^5	1.93×10^6	0.57459	1,637,819.5 $\pm$ 1.42×10^6	0.57459
10^2	4,808.0 $\pm$ 1,868.6	1.09×10^4	0.02271	41,642.1 $\pm$ 31,007.1	1.77×10^5	0.04256	463,857.8 $\pm$ 7.90×10^5	6.76×10^6	0.14564	3,090,607.6 $\pm$ 8.03×10^6	0.14564
10^3	5,079.2 $\pm$ 2,502.6	3.15×10^4	0.00621	40,783.3 $\pm$ 34,004.7	2.51×10^5	0.00616	385,896.2 $\pm$ 7.23×10^5	1.26×10^7	0.03273	3,377,123.9 $\pm$ 6.93×10^6	0.03273
10^4	5,074.5 $\pm$ 2,438.9	3.15×10^4	0.00062	40,390.6 $\pm$ 36,143.8	7.15×10^5	0.00177	365,727.5 $\pm$ 6.22×10^5	2.78×10^7	0.00760	3,498,414.8 $\pm$ 9.81×10^6	0.00760
10^5	5,032.1 $\pm$ 2,436.4	3.31×10^4	0.00007	40,330.1 $\pm$ 34,894.1	8.50×10^5	0.00021	366,894.1 $\pm$ 5.96×10^5	2.89×10^7	0.00079	3,600,385.9 $\pm$ 1.21×10^7	0.00079
10^6	5,038.3 $\pm$ 2,448.0	3.97×10^4	0.00001	40,336.7 $\pm$ 35,324.7	1.18×10^6	0.00003	363,448.7 $\pm$ 5.95×10^5	6.57×10^7	0.00018	3,632,867.4 $\pm$ 1.20×10^7	0.00018
	5,040 #visited: avg = 189.7, max = 194 #generated: avg = 559.8 #saved: avg = 132.3 = 23.6%			40,320 #visited: avg = 260.3, max = 266 #generated: avg = 893.2 #saved: avg = 189.6 = 21.2%			362,880 #visited: avg = 341.1, max = 348 #generated: avg = 1329.5 #saved: avg = 256.0 = 19.3%			3,628,800 #visited: avg = 432.0, max = 439 #generated: avg = 1879.3 #saved: avg = 329.1 = 17.5%	

Permutations										
ax. $\hat{\chi}$		$n = 10$			$n = 11$			$n = 12$		
		mean $\pm$ std. dev.	max.	$\hat{\chi}$	mean $\pm$ std. dev.	max.	$\hat{\chi}$	mean $\pm$ std. dev.	max.	$\hat{\chi}$
$\times 10^6$	0.57459	1,637,819.5 $\pm$ 1.42×10^6	4.70×10^6	0.28716	22,378,645.4 $\pm$ 1.63×10^7	4.65×10^7	0.20799	41,459,073.8 $\pm$ 6.20×10^7	2.02×10^8	0.48692
$\times 10^6$	0.14564	3,090,607.6 $\pm$ 8.03×10^6	7.37×10^7	0.23847	38,544,454.5 $\pm$ 9.60×10^7	7.88×10^8	0.20448	301,450,190.7 $\pm$ 8.95×10^8	5.50×10^9	0.18240
$\times 10^7$	0.03273	3,377,123.9 $\pm$ 6.93×10^6	7.94×10^7	0.02352	48,943,236.2 $\pm$ 2.45×10^8	5.93×10^9	0.12117	323,548,951.4 $\pm$ 1.28×10^9	2.70×10^{10}	0.08359
$\times 10^7$	0.00760	3,498,414.8 $\pm$ 9.81×10^6	4.11×10^8	0.01174	40,249,411.5 $\pm$ 2.51×10^8	1.61×10^{10}	0.03996	881,108,708.2 $\pm$ 4.42×10^{10}	4.40×10^{12}	0.49892
$\times 10^7$	0.00079	3,600,385.9 $\pm$ 1.21×10^7	1.62×10^9	0.00451	38,882,958.2 $\pm$ 2.01×10^8	1.81×10^{10}	0.00465	543,439,948.4 $\pm$ 1.95×10^{10}	4.40×10^{12}	0.08089
$\times 10^7$	0.00018	3,632,867.4 $\pm$ 1.20×10^7	2.25×10^9	0.00062	39,996,802.8 $\pm$ 3.20×10^8	1.30×10^{11}	0.00325	475,854,689.9 $\pm$ 9.59×10^9	4.40×10^{12}	0.00924
= 348		3,628,800 #visited: avg = 432.0, max = 439 #generated: avg = 1879.3 #saved: avg = 329.1 = 17.5%			39,916,800 #visited: avg = 533.0, max = 540 #generated: avg = 2552.7 #saved: avg = 408.6 = 16.0%			479,001,600 #visited: avg = 643.0, max = 650 #generated: avg = 3349.4 #saved: avg = 485.9 = 14.5%		

Table 4: Crowd Sampling, permutations, budget $B = 10$

Klee–Minty cubes											
m	$n = 8$			$n = 10$			$n = 12$			$n = 14$	
	mean $\pm$ std. dev.	max.	$\hat{\chi}$	mean $\pm$ std. dev.	max.	$\hat{\chi}$	mean $\pm$ std. dev.	max.	$\hat{\chi}$	mean $\pm$ std. dev.	max.
10^1	51.2 ± 48.1	1.54×10^2	0.30053	$882.1 \pm 2,154.3$	6.96×10^3	0.78904	$1,046.4 \pm 2,134.4$	6.81×10^3	0.65084	149.9 ± 283.9	9.51×10^2
10^2	106.8 ± 319.3	2.30×10^3	0.21534	181.2 ± 717.4	6.96×10^3	0.38409	$478.4 \pm 1,484.8$	1.21×10^4	0.25341	$1,216.6 \pm 6,959.3$	6.70×10^4
10^3	$167.9 \pm 1,151.1$	2.63×10^4	0.15668	$961.4 \pm 11,280.5$	2.85×10^5	0.29669	$1,787.6 \pm 24,370.0$	6.97×10^5	0.38967	$12,466.1 \pm 3.45 \times 10^5$	1.09×10^7
10^4	$222.6 \pm 3,065.0$	2.03×10^5	0.09107	$711.4 \pm 18,195.4$	1.69×10^6	0.23742	$1,118.3 \pm 16,573.2$	1.05×10^6	0.09420	$3,976.9 \pm 1.28 \times 10^5$	1.09×10^7
10^5	$224.5 \pm 3,794.4$	6.31×10^5	0.02813	$1,146.6 \pm 1.42 \times 10^5$	4.42×10^7	0.38589	$1,499.9 \pm 36,213.5$	5.76×10^6	0.03842	$8,485.1 \pm 1.08 \times 10^6$	3.31×10^8
10^6	$263.5 \pm 27,861.5$	2.70×10^7	0.10262	$854.8 \pm 65,844.1$	4.42×10^7	0.05176	$2,649.3 \pm 3.50 \times 10^5$	2.71×10^8	0.10227	$5,718.6 \pm 6.29 \times 10^5$	4.18×10^8
10^7	$294.9 \pm 1.14 \times 10^5$	3.43×10^8	0.11622	$866.6 \pm 85,253.1$	1.02×10^8	0.01178	$3,542.4 \pm 2.27 \times 10^6$	6.58×10^9	0.18583	$11,649.1 \pm 1.06 \times 10^7$	2.45×10^{10}
	$256 \pm 1.362 \times 10^5$ #visited: avg = 11.5, max = 58			$1,024 \pm 3.246 \times 10^9$ #visited: avg = 15.7, max = 78			$4,096 \pm 1.034 \times 10^{19}$ #visited: avg = 20.6, max = 94			$16,384 \pm 2.024 \times 10^{39}$ #visited: avg = 26.4, max = 1	

Klee–Minty cubes											
$n = 12$			$n = 14$			$n = 16$			$n = 18$		
dev.	max.	$\hat{\chi}$	mean $\pm$ std. dev.	max.	$\hat{\chi}$	mean $\pm$ std. dev.	max.	$\hat{\chi}$	mean $\pm$ std. dev.	max.	$\hat{\chi}$
34.4	6.81×10^3	0.65084	149.9 ± 283.9	9.51×10^2	0.63434	610.8 ± 976.4	2.64×10^3	0.43298	$38,491.9 \pm 1.21 \times 10^5$	3.84×10^5	0.99763
34.8	1.21×10^4	0.25341	$1,216.6 \pm 6,959.3$	6.70×10^4	0.55065	$634.3 \pm 2,131.9$	1.91×10^4	0.30112	$46,731.8 \pm 4.15 \times 10^5$	4.14×10^6	0.88517
370.0	6.97×10^5	0.38967	$12,466.1 \pm 3.45 \times 10^5$	1.09×10^7	0.87548	$3,389.8 \pm 42,683.1$	1.05×10^6	0.31063	$7,316.6 \pm 1.35 \times 10^5$	4.14×10^6	0.56536
373.2	1.05×10^6	0.09420	$3,976.9 \pm 1.28 \times 10^5$	1.09×10^7	0.27443	$9,903.8 \pm 5.95 \times 10^5$	5.92×10^7	0.59781	$25,870.1 \pm 1.17 \times 10^6$	1.03×10^8	0.39674
313.5	5.76×10^6	0.03842	$8,485.1 \pm 1.08 \times 10^6$	3.31×10^8	0.39048	$12,478.0 \pm 1.50 \times 10^6$	4.64×10^8	0.37166	$15,370.4 \pm 7.22 \times 10^5$	1.44×10^8	0.09360
3×10^5	2.71×10^8	0.10227	$5,718.6 \pm 6.29 \times 10^5$	4.18×10^8	0.07303	$17,343.6 \pm 3.44 \times 10^6$	2.55×10^9	0.14700	$57,531.5 \pm 2.94 \times 10^7$	2.91×10^{10}	0.50640
3×10^6	6.58×10^9	0.18583	$11,649.1 \pm 1.06 \times 10^7$	2.45×10^{10}	0.21000	$20,240.1 \pm 6.62 \times 10^6$	9.99×10^9	0.04933	$36,846.6 \pm 1.44 \times 10^7$	2.91×10^{10}	0.07907
34×10^{19} avg = 20.6, max = 94			$16,384 \pm 2.024 \times 10^{39}$ #visited: avg = 26.4, max = 122			$65,536 \pm 3.055 \times 10^{83}$ #visited: avg = 33.0, max = 146			$262,144$ #visited: avg = 40.5, max = 186		

Table 5: Algorithm P, Path Sampling, Klee-Minty cubes

Klee-Minty cubes											
m	$n = 8$			$n = 10$			$n = 12$			$n = 14$	
	mean $\pm$ std. dev.	max.	$\hat{\chi}$	mean $\pm$ std. dev.	max.	$\hat{\chi}$	mean $\pm$ std. dev.	max.	$\hat{\chi}$	mean $\pm$ std. dev.	max.
10^1	244.2 ± 279.1	7.84×10^2	0.32100	87.5 ± 102.1	3.34×10^2	0.38153	$2,028.6 \pm 3,386.0$	1.02×10^4	0.50207	$20,353.3 \pm 62,038.7$	1.97×10^5
10^2	179.0 ± 327.1	2.05×10^3	0.11437	358.4 ± 995.7	7.46×10^3	0.20824	$2,533.0 \pm 13,646.2$	1.33×10^5	0.52576	$3,399.2 \pm 20,190.9$	1.97×10^5
10^3	$324.7 \pm 2,380.0$	6.16×10^4	0.18957	$649.8 \pm 3,521.7$	9.34×10^4	0.14378	$1,699.2 \pm 9,894.0$	1.79×10^5	0.10549	$3,867.5 \pm 22,682.9$	3.61×10^5
10^4	$265.6 \pm 1,581.8$	7.57×10^4	0.02851	$904.4 \pm 24,382.4$	2.38×10^6	0.26310	$2,084.0 \pm 30,840.8$	2.62×10^6	0.12584	$9,624.3 \pm 3.37 \times 10^5$	3.09×10^7
10^5	$242.6 \pm 1,789.8$	3.29×10^5	0.01356	$863.3 \pm 15,913.2$	2.91×10^6	0.03368	$2,686.7 \pm 62,038.0$	1.05×10^7	0.03895	$9,999.6 \pm 6.74 \times 10^5$	1.96×10^8
10^6	$255.0 \pm 3,233.3$	1.23×10^6	0.00482	$963.2 \pm 33,681.1$	1.74×10^7	0.01808	$4,501.2 \pm 8.53 \times 10^5$	6.49×10^8	0.14407	$12,257.7 \pm 1.74 \times 10^6$	1.28×10^9
10^7	$253.0 \pm 3,295.1$	2.92×10^6	0.00115	$986.2 \pm 58,292.5$	9.65×10^7	0.00979	$4,392.0 \pm 1.20 \times 10^6$	3.00×10^9	0.06841	$18,146.7 \pm 1.99 \times 10^7$	6.25×10^{10}
	$256 \pm 6.559 \times 10^3$ #visited: avg = 16.4, max = 56			$1,024 \pm 2.141 \times 10^6$ #visited: avg = 22.4, max = 80			$4,096 \pm 7.762 \times 10^{10}$ #visited: avg = 29.2, max = 104			$16,384 \pm 5.948 \times 10^{19}$ #visited: avg = 36.8, max = 134	

Klee-Minty cubes											
$n = 12$			$n = 14$			$n = 16$			$n = 18$		
dev.	max.	$\hat{\chi}$	mean $\pm$ std. dev.	max.	$\hat{\chi}$	mean $\pm$ std. dev.	max.	$\hat{\chi}$	mean $\pm$ std. dev.	max.	$\hat{\chi}$
3.0	1.02×10^4	0.50207	$20,353.3 \pm 62,038.7$	1.97×10^5	0.96738	$631.4 \pm 1,148.1$	3.76×10^3	0.59521	$42,844.1 \pm 1.16 \times 10^5$	3.70×10^5	0.86268
16.2	1.33×10^5	0.52576	$3,399.2 \pm 20,190.9$	1.97×10^5	0.57924	$3,883.2 \pm 13,555.7$	1.01×10^5	0.25965	$7,680.2 \pm 40,710.0$	3.70×10^5	0.48125
4.0	1.79×10^5	0.10549	$3,867.5 \pm 22,682.9$	3.61×10^5	0.09327	$4,723.2 \pm 27,433.7$	5.29×10^5	0.11204	$34,482.7 \pm 4.17 \times 10^5$	1.03×10^7	0.29867
10.8	2.62×10^6	0.12584	$9,624.3 \pm 3.37 \times 10^5$	3.09×10^7	0.32113	$16,757.6 \pm 4.02 \times 10^5$	2.38×10^7	0.14178	$38,883.5 \pm 1.52 \times 10^6$	1.46×10^8	0.37629
38.0	1.05×10^7	0.03895	$9,999.6 \pm 6.74 \times 10^5$	1.96×10^8	0.19576	$17,017.6 \pm 5.76 \times 10^5$	9.97×10^7	0.05856	$60,680.0 \pm 4.23 \times 10^6$	9.07×10^8	0.14954
$\times 10^5$	6.49×10^8	0.14407	$12,257.7 \pm 1.74 \times 10^6$	1.28×10^9	0.10415	$27,550.0 \pm 3.00 \times 10^6$	1.83×10^9	0.06640	$123,796.1 \pm 5.32 \times 10^7$	5.18×10^{10}	0.41829
$\times 10^6$	3.00×10^9	0.06841	$18,146.7 \pm 1.99 \times 10^7$	6.25×10^{10}	0.34458	$44,123.0 \pm 3.05 \times 10^7$	8.79×10^{10}	0.19929	$99,272.4 \pm 3.38 \times 10^7$	5.65×10^{10}	0.05687
2×10^{10} = 29.2, max = 104			$16,384 \pm 5.948 \times 10^{19}$ #visited: avg = 36.8, max = 134			$65,536 \pm 6.361 \times 10^{36}$ #visited: avg = 45.3, max = 162			$262,144 \pm 1.641 \times 10^{69}$ #visited: avg = 54.6, max = 200		

Table 6: IS by outdegree, Klee-Minty cubes

Klee–Minty cubes											
m	$n = 8$			$n = 10$			$n = 12$			$n = 14$	
	mean $\pm$ std. dev.	max.	$\hat{\chi}$	mean $\pm$ std. dev.	max.	$\hat{\chi}$	mean $\pm$ std. dev.	max.	$\hat{\chi}$	mean $\pm$ std. dev.	max.
10^1	128.2 ± 227.9	6.11×10^2	0.47683	202.2 ± 217.4	6.97×10^2	0.34488	$994.4 \pm 1,564.7$	4.56×10^3	0.45878	550.8 ± 833.1	2.28×10^3
10^2	$338.4 \pm 1,813.0$	1.78×10^4	0.52708	$473.9 \pm 1,054.7$	9.21×10^3	0.19432	$1,171.3 \pm 3,682.9$	3.44×10^4	0.29384	$2,692.4 \pm 18,790.9$	1.86×10^5
10^3	$371.7 \pm 4,890.8$	1.53×10^5	0.41106	$976.3 \pm 10,248.4$	2.90×10^5	0.29717	$2,154.3 \pm 16,630.3$	4.04×10^5	0.18755	$5,845.5 \pm 45,297.2$	8.15×10^5
10^4	$255.5 \pm 2,084.7$	1.53×10^5	0.05981	$1,121.8 \pm 30,028.1$	2.82×10^6	0.25136	$2,337.7 \pm 30,091.7$	2.26×10^6	0.09668	$5,601.6 \pm 57,003.6$	2.43×10^6
10^5	$263.7 \pm 2,767.7$	4.96×10^5	0.01879	$897.2 \pm 13,521.5$	2.82×10^6	0.03143	$3,107.8 \pm 98,543.9$	2.37×10^7	0.07613	$9,261.9 \pm 3.52 \times 10^5$	6.18×10^7
10^6	$257.8 \pm 3,490.3$	1.24×10^6	0.00480	$1,026.2 \pm 61,065.3$	4.95×10^7	0.04828	$3,255.6 \pm 1.65 \times 10^5$	7.04×10^7	0.02163	$42,306.9 \pm 2.41 \times 10^7$	2.27×10^{10}
10^7	$256.5 \pm 4,163.8$	5.93×10^6	0.00231	$1,006.2 \pm 65,644.4$	1.29×10^8	0.01282	$3,635.2 \pm 1.01 \times 10^6$	3.11×10^9	0.08555	$17,241.0 \pm 9.02 \times 10^6$	2.27×10^{10}
	$256 \pm 5.045 \times 10^3$ #visited: avg = 16.4, max = 60			$1,024 \pm 2.080 \times 10^6$ #visited: avg = 22.4, max = 80			$4,096 \pm 6.025 \times 10^{11}$ #visited: avg = 29.2, max = 112			$16,384 \pm 1.718 \times 10^{24}$ #visited: avg = 36.8, max = 138	

Klee–Minty cubes											
$n = 12$			$n = 14$			$n = 16$			$n = 18$		
dev.	max.	$\hat{\chi}$	mean $\pm$ std. dev.	max.	$\hat{\chi}$	mean $\pm$ std. dev.	max.	$\hat{\chi}$	mean $\pm$ std. dev.	max.	$\hat{\chi}$
7	4.56×10^3	0.45878	550.8 ± 833.1	2.28×10^3	0.41460	$1,211.8 \pm 1,761.6$	5.11×10^3	0.42200	$3,435.1 \pm 8,094.7$	2.62×10^4	0.76167
9	3.44×10^4	0.29384	$2,692.4 \pm 18,790.9$	1.86×10^5	0.69180	$2,910.7 \pm 10,412.5$	9.31×10^4	0.31972	$9,823.5 \pm 30,415.3$	2.01×10^5	0.20502
0.3	4.04×10^5	0.18755	$5,845.5 \pm 45,297.2$	8.15×10^5	0.13949	$10,251.5 \pm 1.38 \times 10^5$	3.73×10^6	0.36387	$8,205.0 \pm 53,192.3$	1.10×10^6	0.13419
1.7	2.26×10^6	0.09668	$5,601.6 \pm 57,003.6$	2.43×10^6	0.04342	$10,447.2 \pm 1.23 \times 10^5$	5.94×10^6	0.05687	$26,703.2 \pm 6.55 \times 10^5$	4.98×10^7	0.18635
3.9	2.37×10^7	0.07613	$9,261.9 \pm 3.52 \times 10^5$	6.18×10^7	0.06670	$41,666.4 \pm 6.35 \times 10^6$	1.98×10^9	0.47420	$50,077.2 \pm 2.70 \times 10^6$	4.90×10^8	0.09783
10^5	7.04×10^7	0.02163	$42,306.9 \pm 2.41 \times 10^7$	2.27×10^{10}	0.53708	$118,559.5 \pm 9.22 \times 10^7$	9.21×10^{10}	0.77684	$116,464.2 \pm 4.42 \times 10^7$	4.31×10^{10}	0.37028
10^6	3.11×10^9	0.08555	$17,241.0 \pm 9.02 \times 10^6$	2.27×10^{10}	0.13179	$44,136.5 \pm 3.10 \times 10^7$	9.21×10^{10}	0.20868	$157,239.5 \pm 2.01 \times 10^8$	6.29×10^{11}	0.39984
$\times 10^{11}$ = 29.2, max = 112			$16,384 \pm 1.718 \times 10^{24}$ #visited: avg = 36.8, max = 138			$65,536 \pm 3.318 \times 10^{51}$ #visited: avg = 45.3, max = 158			$262,144 \pm 4.872 \times 10^{113}$ #visited: avg = 54.6, max = 192		

Table 7: IS by size, Klee–Minty cubes

Klee–Minty cubes											
m	$n = 8$			$n = 10$			$n = 12$			$n = 14$	
	mean $\pm$ std. dev.	max.	$\hat{\chi}$	mean $\pm$ std. dev.	max.	$\hat{\chi}$	mean $\pm$ std. dev.	max.	$\hat{\chi}$	mean $\pm$ std. dev.	max.
10^1	208.6 ± 192.1	7.20×10^2	0.34514	583.2 ± 611.6	2.07×10^3	0.35412	$2,778.3 \pm 5,045.8$	1.67×10^4	0.60117	$7,859.3 \pm 12,982.6$	4.06×10^4
10^2	301.1 ± 642.9	6.13×10^3	0.20370	$1,299.8 \pm 4,881.0$	4.77×10^4	0.36690	$2,231.1 \pm 5,569.1$	4.60×10^4	0.20638	$12,987.3 \pm 39,685.0$	2.82×10^5
10^3	261.2 ± 438.0	6.13×10^3	0.02348	$897.7 \pm 2,388.1$	4.77×10^4	0.05313	$4,351.1 \pm 55,366.3$	1.74×10^6	0.39894	$27,133.5 \pm 2.90 \times 10^5$	7.45×10^6
10^4	251.8 ± 435.5	8.64×10^3	0.00343	$976.6 \pm 3,751.7$	1.52×10^5	0.01551	$3,611.7 \pm 46,923.6$	4.00×10^6	0.11076	$14,050.0 \pm 1.66 \times 10^5$	7.46×10^6
10^5	255.5 ± 520.0	2.94×10^4	0.00115	$1,066.5 \pm 19,675.0$	5.92×10^6	0.05552	$3,851.3 \pm 58,920.5$	1.00×10^7	0.02597	$17,321.6 \pm 5.94 \times 10^5$	1.33×10^8
10^6	256.5 ± 536.5	7.53×10^4	0.00029	$1,034.9 \pm 12,622.9$	6.65×10^6	0.00643	$4,031.7 \pm 1.21 \times 10^5$	7.25×10^7	0.01799	$14,878.3 \pm 6.18 \times 10^5$	3.76×10^8
10^7	256.0 ± 536.9	1.54×10^5	0.00006	$1,023.2 \pm 8,554.4$	9.69×10^6	0.00095	$4,043.8 \pm 1.38 \times 10^5$	2.93×10^8	0.00725	$14,986.2 \pm 7.80 \times 10^5$	1.13×10^9
	$256 \pm 5.394 \times 10^2$ #visited: avg = 28.9, max = 78			$1,024 \pm 1.071 \times 10^4$ #visited: avg = 42.5, max = 112			$4,096 \pm 9.419 \times 10^5$ #visited: avg = 58.7, max = 160			$16,384 \pm 1.692 \times 10^9$ #visited: avg = 77.6, max = 204	

Klee–Minty cubes											
$n = 12$			$n = 14$			$n = 16$			$n = 18$		
dev.	max.	$\hat{\chi}$	mean $\pm$ std. dev.	max.	$\hat{\chi}$	mean $\pm$ std. dev.	max.	$\hat{\chi}$	mean $\pm$ std. dev.	max.	$\hat{\chi}$
5.8	1.67×10^4	0.60117	$7,859.3 \pm 12,982.6$	4.06×10^4	0.51720	$7,026.3 \pm 6,937.8$	2.34×10^4	0.33300	$449,457.9 \pm 1.38 \times 10^6$	4.38×10^6	0.97383
9.1	4.60×10^4	0.20638	$12,987.3 \pm 39,685.0$	2.82×10^5	0.21702	$21,245.7 \pm 1.00 \times 10^5$	9.60×10^5	0.45163	$79,455.3 \pm 4.59 \times 10^5$	4.38×10^6	0.55087
66.3	1.74×10^6	0.39894	$27,133.5 \pm 2.90 \times 10^5$	7.45×10^6	0.27456	$72,946.6 \pm 1.39 \times 10^6$	4.31×10^7	0.59144	$59,564.7 \pm 3.98 \times 10^5$	9.29×10^6	0.15597
23.6	4.00×10^6	0.11076	$14,050.0 \pm 1.66 \times 10^5$	7.46×10^6	0.05309	$51,458.3 \pm 1.46 \times 10^6$	1.34×10^8	0.25977	$179,949.2 \pm 6.40 \times 10^6$	6.00×10^8	0.33317
20.5	1.00×10^7	0.02597	$17,321.6 \pm 5.94 \times 10^5$	1.33×10^8	0.07653	$40,430.8 \pm 8.29 \times 10^5$	1.34×10^8	0.03306	$161,072.6 \pm 1.05 \times 10^7$	3.00×10^9	0.18601
$\times 10^5$	7.25×10^7	0.01799	$14,878.3 \pm 6.18 \times 10^5$	3.76×10^8	0.02525	$48,306.2 \pm 2.61 \times 10^6$	1.03×10^9	0.02125	$206,329.3 \pm 2.81 \times 10^7$	2.07×10^{10}	0.10042
$\times 10^5$	2.93×10^8	0.00725	$14,986.2 \pm 7.80 \times 10^5$	1.13×10^9	0.00752	$80,786.9 \pm 8.83 \times 10^7$	2.78×10^{11}	0.34431	$171,868.4 \pm 2.00 \times 10^7$	2.55×10^{10}	0.01484
9×10^5 avg = 58.7, max = 160			$16,384 \pm 1.692 \times 10^9$ #visited: avg = 77.6, max = 204			$65,536 \pm 1.501 \times 10^{15}$ #visited: avg = 99.2, max = 248			$262,144 \pm 2.711 \times 10^{26}$ #visited: avg = 123.6, max = 310		

Table 8: IS by size/indegree, Klee–Minty cubes

Klee–Minty cubes											
m	$n = 8$			$n = 10$			$n = 12$			$n = 14$	
	mean $\pm$ std. dev.	max.	$\hat{\chi}$	mean $\pm$ std. dev.	max.	$\hat{\chi}$	mean $\pm$ std. dev.	max.	$\hat{\chi}$	mean $\pm$ std. dev.	max.
10^1	160.5 ± 90.1	3.32×10^2	0.20676	$1,183.2 \pm 1,423.3$	4.64×10^3	0.39191	$2,930.8 \pm 2,237.0$	7.84×10^3	0.26759	$5,761.1 \pm 4,945.3$	1.30×10^4
10^2	196.6 ± 162.9	9.08×10^2	0.04618	$1,171.8 \pm 2,521.9$	1.91×10^4	0.16258	$3,486.7 \pm 5,502.1$	3.70×10^4	0.10600	$25,459.0 \pm 1.47 \times 10^5$	1.44×10^6
10^3	243.0 ± 357.9	7.11×10^3	0.02925	$1,111.9 \pm 3,284.9$	7.72×10^4	0.06940	$3,711.8 \pm 13,201.9$	2.79×10^5	0.07510	$15,876.8 \pm 84,660.2$	1.91×10^6
10^4	255.4 ± 395.3	1.51×10^4	0.00591	$1,022.8 \pm 3,526.0$	2.42×10^5	0.02371	$3,964.3 \pm 18,223.7$	1.15×10^6	0.02899	$13,916.5 \pm 59,069.0$	2.10×10^6
10^5	256.4 ± 391.3	1.83×10^4	0.00072	$1,014.0 \pm 3,282.6$	3.22×10^5	0.00318	$3,993.5 \pm 27,014.1$	4.98×10^6	0.01246	$15,669.2 \pm 1.94 \times 10^5$	3.36×10^7
10^6	255.7 ± 392.1	5.93×10^4	0.00023	$1,031.1 \pm 4,959.5$	2.97×10^6	0.00288	$4,104.4 \pm 44,370.2$	2.05×10^7	0.00500	$17,741.3 \pm 1.51 \times 10^6$	1.48×10^9
	256 #visited: avg = 74.1, max = 329 #generated: avg = 293.7			1,024 #visited: avg = 193.2, max = 1328 #generated: avg = 936.6			4,096 #visited: avg = 543.1, max = 4889 #generated: avg = 3096.1			16,384 #visited: avg = 1674.3, max = 15202 #generated: avg = 10954.4	

Klee–Minty cubes											
$n = 12$			$n = 14$			$n = 16$			$n = 18$		
std. dev.	max.	$\hat{\chi}$	mean $\pm$ std. dev.	max.	$\hat{\chi}$	mean $\pm$ std. dev.	max.	$\hat{\chi}$	mean $\pm$ std. dev.	max.	$\hat{\chi}$
$2,237.0$	7.84×10^3	0.26759	$5,761.1 \pm 4,945.3$	1.30×10^4	0.22622	$15,257.2 \pm 16,978.9$	6.11×10^4	0.40043	$31,338.8 \pm 36,229.9$	1.25×10^5	0.39944
$5,502.1$	3.70×10^4	0.10600	$25,459.0 \pm 1.47 \times 10^5$	1.44×10^6	0.56428	$39,865.8 \pm 1.14 \times 10^5$	1.01×10^6	0.25373	$135,710.8 \pm 3.90 \times 10^5$	2.98×10^6	0.21982
$13,201.9$	2.79×10^5	0.07510	$15,876.8 \pm 84,660.2$	1.91×10^6	0.12042	$50,342.5 \pm 2.41 \times 10^5$	5.77×10^6	0.11457	$186,049.0 \pm 1.28 \times 10^6$	3.71×10^7	0.19957
$18,223.7$	1.15×10^6	0.02899	$13,916.5 \pm 59,069.0$	2.10×10^6	0.01512	$56,098.6 \pm 7.37 \times 10^5$	6.58×10^7	0.11736	$238,368.8 \pm 5.68 \times 10^6$	5.47×10^8	0.22944
$27,014.1$	4.98×10^6	0.01246	$15,669.2 \pm 1.94 \times 10^5$	3.36×10^7	0.02142	$67,285.8 \pm 1.45 \times 10^6$	2.65×10^8	0.03932	$257,051.0 \pm 1.72 \times 10^7$	5.34×10^9	0.20793
$44,370.2$	2.05×10^7	0.00500	$17,741.3 \pm 1.51 \times 10^6$	1.48×10^9	0.08341	$63,947.2 \pm 2.75 \times 10^6$	2.08×10^9	0.03248	$231,119.3 \pm 8.31 \times 10^6$	5.34×10^9	0.02313
avg = 543.1, max = 4889 std: avg = 3096.1			16,384 #visited: avg = 1674.3, max = 15202 #generated: avg = 10954.4			65,536 #visited: avg = 5771.8, max = 66585 #generated: avg = 42684.7			262,144 #visited: avg = 22389.6, max = 305716 #generated: avg = 185115.0		

Table 9: Bucket Sampling, Klee–Minty cubes

Klee–Minty cubes											
m	$n = 8$			$n = 10$			$n = 12$			$n = 14$	
	mean $\pm$ std. dev.	max.	$\hat{\chi}$	mean $\pm$ std. dev.	max.	$\hat{\chi}$	mean $\pm$ std. dev.	max.	$\hat{\chi}$	mean $\pm$ std. dev.	max.
10^1	238.4 ± 111.7	4.33×10^2	0.18184	$1,508.5 \pm 1,520.0$	4.91×10^3	0.32530	$2,741.1 \pm 2,785.5$	9.82×10^3	0.35839	$8,289.1 \pm 15,826.3$	5.26×10^4
10^2	285.7 ± 307.8	2.67×10^3	0.09347	939.6 ± 998.9	5.31×10^3	0.05647	$4,860.2 \pm 14,275.1$	1.38×10^5	0.28425	$12,516.4 \pm 21,836.3$	1.36×10^5
10^3	257.5 ± 195.2	2.67×10^3	0.01037	$1,028.3 \pm 1,292.8$	1.62×10^4	0.01578	$3,988.3 \pm 8,651.3$	1.38×10^5	0.03464	$13,986.3 \pm 35,657.4$	5.66×10^5
10^4	252.9 ± 178.9	2.67×10^3	0.00106	$1,001.7 \pm 1,342.2$	4.10×10^4	0.00409	$4,124.8 \pm 11,867.5$	6.98×10^5	0.01692	$16,142.2 \pm 76,318.2$	3.08×10^6
10^5	255.2 ± 181.0	4.55×10^3	0.00018	$1,018.4 \pm 1,442.2$	6.95×10^4	0.00068	$4,069.6 \pm 12,271.1$	1.07×10^6	0.00263	$16,842.5 \pm 1.77 \times 10^5$	3.66×10^7
10^6	255.9 ± 180.3	7.24×10^3	0.00003	$1,024.6 \pm 1,483.7$	1.28×10^5	0.00012	$4,082.1 \pm 13,309.8$	3.93×10^6	0.00096	$16,266.9 \pm 1.41 \times 10^5$	4.97×10^7
	256 #visited: avg = 490.8, max = 939 #generated: avg = 1405.8 #saved: avg = 343.5 = 24.4%			1,024 #visited: avg = 709.3, max = 1441 #generated: avg = 2486.1 #saved: avg = 392.2 = 15.8%			4,096 #visited: avg = 937.7, max = 1839 #generated: avg = 3892.1 #saved: avg = 416.2 = 10.7%			16,384 #visited: avg = 1179.0, max = 2340 #generated: avg = 5673.0 #saved: avg = 433.8 = 7.6%	

Klee–Minty cubes											
$n = 12$			$n = 14$			$n = 16$			$n = 18$		
std. dev.	max.	$\hat{\chi}$	mean $\pm$ std. dev.	max.	$\hat{\chi}$	mean $\pm$ std. dev.	max.	$\hat{\chi}$	mean $\pm$ std. dev.	max.	$\hat{\chi}$
2,785.5	9.82×10^3	0.35839	$8,289.1 \pm 15,826.3$	5.26×10^4	0.63447	$13,171.2 \pm 14,840.3$	4.18×10^4	0.31701	$68,493.5 \pm 1.25 \times 10^5$	4.15×10^5	0.60551
4,275.1	1.38×10^5	0.28425	$12,516.4 \pm 21,836.3$	1.36×10^5	0.10896	$122,995.2 \pm 6.83 \times 10^5$	6.68×10^6	0.54346	$220,869.2 \pm 1.23 \times 10^6$	1.22×10^7	0.55364
8,651.3	1.38×10^5	0.03464	$13,986.3 \pm 35,657.4$	5.66×10^5	0.04045	$59,153.4 \pm 2.90 \times 10^5$	6.68×10^6	0.11300	$178,839.3 \pm 1.15 \times 10^6$	2.91×10^7	0.16283
11,867.5	6.98×10^5	0.01692	$16,142.2 \pm 76,318.2$	3.08×10^6	0.01909	$58,608.2 \pm 5.38 \times 10^5$	3.03×10^7	0.05162	$235,063.5 \pm 4.54 \times 10^6$	3.99×10^8	0.16994
12,271.1	1.07×10^6	0.00263	$16,842.5 \pm 1.77 \times 10^5$	3.66×10^7	0.02175	$65,936.6 \pm 1.82 \times 10^6$	5.34×10^8	0.08096	$268,082.9 \pm 8.11 \times 10^6$	2.18×10^9	0.08132
13,309.8	3.93×10^6	0.00096	$16,266.9 \pm 1.41 \times 10^5$	4.97×10^7	0.00306	$72,790.8 \pm 8.86 \times 10^6$	8.79×10^9	0.12082	$249,829.7 \pm 7.68 \times 10^6$	3.57×10^9	0.01429
avg = 937.7, max = 1839 std: avg = 3892.1 avg = 416.2 = 10.7%			16,384 #visited: avg = 1179.0, max = 2340 #generated: avg = 5673.0 #saved: avg = 433.8 = 7.6%			65,536 #visited: avg = 1431.7, max = 2861 #generated: avg = 7824.9 #saved: avg = 446.0 = 5.7%			262,144 #visited: avg = 1694.7, max = 3421 #generated: avg = 10361.8 #saved: avg = 456.6 = 4.4%		

Table 10: Crowd Sampling, budget $B = 10$, Klee-Minty cubes